Sequential Decision Making: Theory and Practice

Lecture 2: Adversarial Bandits and Lower Bounds

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Outline

Project schedule

It is not too early to start thinking about the project!

The proposal stage: the proposal sets you ready for the actual research: well-defined problem, well-calibrated baseline, well-planned action items...

Timeline:

- Sep. 22: project proposal, first draft
- Oct. 6: project proposal, second draft
- Oct. 13: midterm presentation draft
- Oct. 20: midterm presentation and peer feedback
- Oct. 27: project proposal, final version

Project proposal: first draft

What do we expect from the first draft? brainstorming!

- what?
- why?
- how?
- backup plan?

Page limit for first draft: 2. no smaller than 10 pt font, 1-in margin.

The project should be independent work. But, a group of two is allowed, where the amount of work should be commensurate with the group size.

Recap: stochastic bandit with i.i.d. rewards

We consider a simple setting with i.i.d. bounded rewards.

- Each arm distributes rewards according to some (unknown) distribution over $[0, 1]$, with

$$\mathbb{E}[r_{i,t}] = \mu_i, \quad \forall i \in [n],\ t = 1, 2 \ldots$$

- Suppose we play arm i_t at round t, and receive the reward

$$r_{i_t,t}$$

drawn i.i.d. from the arm i_t's distribution.

Partial information: Every round we cannot observe the reward of all arms: we just know the reward of the arm that we played.

Regret: performance metric

We design algorithms that determine the sequence $\{i_t\}$, i.e. *policies*.

How to evaluate the performance?

Definition 1 (Expected regret)

The expected regret over T rounds is defined as

$$R_T = \max_{1 \le i \le n} \mathbb{E}\left[\sum_{t=1}^{T} (r_{i,t} - r_{i_t,t})\right] = T\mu^\star - \mathbb{E}\left[\sum_{t=1}^{T} r_{i_t,t}\right],$$

where $\mu^\star = \max_{1 \le i \le n} \mu_i$ is the highest expected reward over all arms.

- 1st term captures the highest cumulative reward in *hindsight*.
- 2nd term captures the *actual* accumulated reward.

The UCB algorithm

[Auer et al., 2002a]: the idea is to always try the best arm, where "best" includes exploration and exploitation.

1. **Initial phase:** try each arm and observe the reward.
2. For each round $t = n+1, \ldots, T$:
   - Calculate the UCB (upper confidence bound) index for each arm i:

$$\mathsf{UCB}_{i,t} = \overline{\mu}_{i,t} + \sqrt{\frac{\log t}{T_{i,t}}},$$

     where $\overline{\mu}_{i,t}$ is the empirical average reward for arm i and $T_{i,t}$ is the number of times arm i has been played up to round t.

   - Play the arm with the highest UCB index and observe the reward.

Theory of UCB algorithm

Theorem 2 (Instance-dependent regret bound of UCB)

For $T \geq n$, the expected regret of UCB algorithm is upper bounded as

$$R_T \leq \sum_{i:\Delta_i>0} \left(\frac{4 \log T}{\Delta_i} + 8\Delta_i \right) \leq \sum_{i:\Delta_i>0} \frac{4 \log T}{\Delta_i} + 8n,$$

where $\Delta_i = \mu^\star - \mu_i$ is the sub-optimality gap of arm i.

Theorem 3 (Instance-independent regret bound of UCB)

For $T \geq n$, the expected regret of UCB algorithm is upper bounded as

$$R_T \leq 4\sqrt{nT \log T} + 8n.$$

- $\left(\sum_{i:\Delta_i>0} \frac{1}{\Delta_i}\right) \log T$ versus $\sqrt{T}$ regret.

Introduction to adversarial bandits

Limitations of stochastic bandits

In stochastic bandits, the reward distributions of the stochastic bandits do not depend on past rewards or actions, and the draws are i.i.d. over time.

This can be too restrictive, and unrealistic in practice.

- Is there a reward distribution?
- Are the rewards i.i.d.?

Adversarial bandits

Idea: make minimal assumptions about reward generation, and compete with the best action in hindsight.

For an n-arm adversarial bandit,

- the rewards are in the interval $[0, 1]$
- in each round t, the reward vector $r_t = [r_{i,t}]_{1 \leq i \leq n}$ is an arbitrary sequence
- all rewards are determined before action is taken

Example

Let $n = 3$, consider the following arbitrary rewards ...

k	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$\ldots$
1	0.1	0.7	0.4	0.3	
2	0.5	0.1	0.6	0.1	
3	0.8	0.4	0.4	0.8	

- If action = arm 1, $\sum_{t=1}^{4} r_{1,t} = 0.1 + 0.7 + 0.4 + 0.3 = 1.5$;
- If action = arm 2, $\sum_{t=1}^{4} r_{2,t} = 0.5 + 0.1 + 0.6 + 0.1 = 1.4$;
- If action = arm 3, $\sum_{t=1}^{4} r_{3,t} = 0.8 + 0.4 + 0.4 + 0.8 = 2.4$;
- The best arm is arm 3.
- The best arm might change with the rewards.

Can we compete with the best arm no matter what the rewards are?

Performance metric: regret

Definition 4 (Expected regret)

The expected regret over T rounds for an action selection rule is defined as

$$R_T(r) = \max_{1 \le i \le n} \sum_{t=1}^{T} r_{i,t} - \sum_{t=1}^{T} \mathbb{E}\left[r_{i_t,t}\right],$$

where the expectation is over the randomness of the learner's actions, and $r := \{r_t\}_{t=1}^{T}$ is the reward. The worst-case regret over all rewards is

$$R_T^\star = \sup_{r \in [0,1]^{T \times n}} R_T(r).$$

- Minimizing the regret has a "min-max" flavor.

Goal: achieve sublinear regret $R_T^\star = o(T)$.

Algorithms: from full-information to bandits

The power of randomization

- Exploration vs exploitation remains an important issue.
- Exploitation appears to be even more dangerous.
  - adversary can “exploit our exploitation”.

Example: Let $n = 3$, the selection of the learner is given in red.

k	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$\dots$
1	0.1	0.7	0	0	
2	0	0.1	0.6	0.1	
3	0.8	0	0.4	0.8	

- For learner, $\sum_{t=1}^{4} r'_{i_t,t} = 0 + 0 + 0 + 0 = 0$.
- $R_4(r') = \max_{i=1,2,3} \sum_{t=1}^{4} r'_{i,t} = 2$. High regret!

The power of randomization

Without randomization in actions, the regret can be linear.

A construction: for any deterministic sequence of actions $\{i_t\}_{t=1}^T$, we can construct a reward sequence such that

$$\forall t : \quad r_{i,t} = \begin{cases} 0 & i = i_t \\ 1 & \text{otherwise} \end{cases}$$

By construction,

$$\sum_{i=1}^{n} \sum_{t=1}^{T} r_{i,t} = (n-1)T \quad \Longrightarrow \max_{1 \le i \le n} \sum_{t=1}^{4} r_{i,t} \ge \frac{(n-1)T}{n}.$$

$$\Longrightarrow R_n(r) = \max_{1 \le i \le n} \sum_{t=1}^{T} r_{i,t} - \underbrace{\sum_{i=1}^{T} r_{i_t,t}}_{=0} \ge \frac{(n-1)T}{n}!$$

Detour: bandits with full information

Randomization: Let $p_t = [p_{i,t}]$ be the probability of choosing different arms in round t.

How to design/update p_t?

Detour: Let's visit the bandit problem with full information, where we observe the entire reward vector

$$r_t = [r_{1,t}, r_{2,t}, \ldots, r_{n,t}].$$

This is also known as *online learning with expert advice*, often formulated with losses rather than rewards. A huge field!

Exponential weight algorithm

Recall the regret

$$R_T(r) = \max_{1 \le i \le n} \sum_{t=1}^{T} \left(r_{i,t} - \mathbb{E}_{i_t \sim p_t} [r_{i_t,t}] \right).$$

Exponential-weight algorithm: assign a higher probability to arms with better performance according to exponential weights [Vovk, 1990, Littlestone and Warmuth, 1994]:

$$p_{i,t+1} \propto \exp\left(\eta \sum_{\ell=1}^{t} r_{i,\ell}\right) = \frac{\exp\left(\eta \sum_{\ell=1}^{t} r_{i,\ell}\right)}{\sum_{i=1}^{n} \exp\left(\eta \sum_{\ell=1}^{t} r_{i,\ell}\right)}, \qquad 1 \le i \le n$$

where $\eta > 0$ is some parameter.

- The randomness ensures exploration: even arms with 0 cumulative rewards so far get chances
- Higher weights encourage exploitation: η controls the trade-off exploitation and exploration

A bit more discussion

Incremental update:

$$p_{i,t+1} \propto \exp\left(\eta \sum_{\ell=1}^{t} r_{i,\ell}\right) \propto p_{i,t} \cdot e^{\eta r_{i,t}}$$

- A multiplicative combination of history information and new information

This algorithm has many names, e.g. multiplicative weight updates (MWU) method, Hedge, and was rediscovered many times in different contexts; see [Arora et al., 2012] for more information.

Exponential weight algorithm with full information

1. **Initialization:** set p_1 as a uniform distribution over $[n]$; set $S_{i,0} = 0$ for $i \in [n]$; parameter $\eta > 0$.
2. For each round $t = 1, 2, \ldots, T$:
   - Observe the reward $r_{i,t}$ for $i \in [n]$, and update the cumulative reward

$$S_{i,t} = S_{i,t-1} + r_{i,t}.$$

   - Update the sampling probability over arms

$$p_{i,t+1} \propto \exp\left(\eta S_{i,t}\right).$$

What is the regret of exponential weight algorithm?

$$R_T(r) = \max_{1 \le i \le n} \sum_{t=1}^{T} \left(r_{i,t} - \mathbb{E}\left[r_{i_t,t}\right]\right) = \max_{1 \le i \le n} \sum_{t=1}^{T} r_{i,t} - \sum_{t=1}^{T} \sum_{i=1}^{n} p_{i,t} r_{i,t}.$$

Regret of exponential weight algorithm

Theorem 5 (Regret under full information)

The exponential weights algorithm with parameter $\eta > 0$ incurs regret

$$R_T^\star = \sup_{r \in [0,1]^{T \times n}} R_T(r) \leq T\eta + \frac{\log n}{\eta}.$$

Choosing $\eta = \sqrt{\log n / T}$ gives

$$R_T^\star \leq 2\sqrt{T \log n}.$$

sublinear regret!

Can we translate this back to the bandit setting?

Back to bandits

In bandits, we only observe $r_{i_t,t}$ for the pulled arm i_t!

A general recipe:

- Step 1: estimate the entire reward vector
- Step 2: plug this into the full-information algorithm

Importance-sampling estimator:

$$\widetilde{r}_{i,t} = \frac{r_{i,t}}{p_{i,t}} \mathbb{I}_{i_t = i} = \begin{cases} \frac{r_{i_t,t}}{p_{i_t,t}} & i = i_t \\ 0 & \text{otherwise} \end{cases}$$

or equivalently

$$\widetilde{r}_t = \left[0, \ldots, \frac{r_{i_t,t}}{p_{i_t,t}}, \ldots, 0 \right].$$

This is an *unbiased* estimate of the reward vector: $\mathbb{E}[\widetilde{r}_{i,t}] = r_{i,t}$.

EXP3 for adversarial bandits

[Auer et al., 2002b]: EXP3 $=$ Exponential-weight algorithm for Exploration and Exploitation

1. **Initialization:** set p_1 as a uniform distribution over $[n]$; set $\widetilde{S}_{i,0} = 0$ for $i \in [n]$; parameter $\eta > 0$.
2. For each round $t = 1, 2, \ldots, T$:
   - Draw an arm i_t from the distribution p_t;
   - For each arm $i \in [n]$, compute the estimated reward

     $$\widetilde{r}_{i,t} = \frac{r_{i,t}}{p_{i,t}} \mathbb{I}_{i_t = i}$$

     and update the cumulative reward $\widetilde{S}_{i,t} = \widetilde{S}_{i,t-1} + \widetilde{r}_{i,t}$.
   - Update the sampling probability over arms

     $$p_{i,t+1} \propto \exp\left(\eta \widetilde{S}_{i,t}\right).$$

Regret of EXP3

Theorem 6 (Regret of EXP3)

The EXP3 algorithm with parameter $\eta > 0$ incurs regret

$$R_T^\star \le nT\eta + \frac{\log n}{\eta}.$$

- The first term is worse by a factor of n compared with the full-information case;
- Choosing $\eta = \sqrt{\log n/(nT)}$ gives

$$R_T^\star \le 2\sqrt{nT\log n}.$$

- Adversarial bandits are not harder than stochastic bandits: matches the worst-case regret bound $\widetilde{O}(\sqrt{nT})$ of UCB for stochastic bandits.

Analysis

Roadmap

- We will first prove the regret bound for the full-information setting.
- We then adapt it to the bandit setting.

Proof of Theorem 2 (full information)

Step 1: introduce the "magic" measure of progress (log-sum-exp):

$$\Phi_t = \frac{1}{\eta} \log \left(\sum_{i=1}^{n} \exp(\eta S_{i,t}) \right)$$

Recall that $p_{i,t+1} = \frac{\exp\left(\eta S_{i,t}\right)}{\sum_{i=1}^{n} \exp\left(\eta S_{i,t}\right)}$.

Basic facts:

- $\Phi_0 = \frac{1}{\eta} \log n$.
- $\Phi_T = \frac{1}{\eta} \log \left(\sum_{i=1}^{n} \exp(\eta S_{i,T}) \right) \geq \frac{1}{\eta} \log \left(\exp(\eta S_{i,T}) \right) = S_{i,T}$, for $i \in [n]$; in other words, Φ_T upper bounds the performance of individual arms, the first term in the regret.

Proof of Theorem 2 (full information)

Step 2: understand the temporal difference of Φ_t.

$$\begin{aligned}
\Phi_t - \Phi_{t-1} &= \frac{1}{\eta}\log\left(\sum_{i=1}^{n} e^{\eta S_{i,t}}\right) - \frac{1}{\eta}\log\left(\sum_{i=1}^{n} e^{\eta S_{i,t-1}}\right) \\
&= \frac{1}{\eta}\log\frac{\sum_{i=1}^{n} e^{\eta S_{i,t-1}} e^{\eta r_{i,t}}}{\sum_{i=1}^{n} e^{\eta S_{i,t-1}}} \\
&= \frac{1}{\eta}\log\sum_{i=1}^{n} \frac{e^{\eta S_{i,t-1}}}{\sum_{i=1}^{n} e^{\eta S_{i,t-1}}} e^{\eta r_{i,t}} \\
&= \frac{1}{\eta}\log\sum_{i=1}^{n} p_{i,t} e^{\eta r_{i,t}},
\end{aligned}$$

which is linked to the sampling probabilities.

Proof of Theorem 2 (full information)

Step 2 - continued: deconstructing the log-sum, using basic facts:

$$
\begin{aligned}
\Phi_t - \Phi_{t-1} &= \frac{1}{\eta} \log \sum_{i=1}^{n} p_{i,t} e^{\eta r_{i,t}} \\
&\leq \frac{1}{\eta} \log \sum_{i=1}^{n} p_{i,t} \left(1 + \eta r_{i,t} + \eta^2 r_{i,t}^2\right) \qquad (e^x \leq 1 + x + x^2) \\
&\leq \frac{1}{\eta} \log \left(1 + \eta \sum_{i=1}^{n} p_{i,t} r_{i,t} + \eta^2 \sum_{i=1}^{n} p_{i,t} r_{i,t}^2\right) \\
&\leq \sum_{i=1}^{n} p_{i,t} r_{i,t} + \eta \sum_{i=1}^{n} p_{i,t} r_{i,t}^2 \qquad (\log(1+x) \leq x)
\end{aligned}
$$

- $\forall x : \quad \log(1+x) \leq x$
- For $x \leq 1 : \quad e^x \leq 1 + x + x^2$

Proof of Theorem 2 (full information)

Step 3: by telescoping

$$\Phi_T - \Phi_0 = \sum_{t=1}^{T}(\Phi_t - \Phi_{t-1}) \le \sum_{t=1}^{T}\sum_{i=1}^{n} p_{i,t} r_{i,t} + \eta \sum_{t=1}^{T}\sum_{i=1}^{n} p_{i,t} r_{i,t}^2.$$

Therefore,

$$\begin{aligned}
R_T(r) = \max_{1\le i\le n} S_{i,T} - \sum_{t=1}^{T}\sum_{i=1}^{n} p_{i,t} r_{i,t} &\le \Phi_T - \sum_{t=1}^{T}\sum_{i=1}^{n} p_{i,t} r_{i,t} \qquad (S_{i,T} \le \Phi_T) \\
&\le \Phi_0 + \eta \sum_{t=1}^{T}\sum_{i=1}^{n} p_{i,t} r_{i,t}^2 \\
&\le \frac{1}{\eta}\log n + \eta \sum_{t=1}^{T}\sum_{i=1}^{n} p_{i,t} \quad \left(\Phi_0 \le \frac{1}{\eta}\log n\right) \\
&\le \frac{\log n}{\eta} + \eta T. \quad \Big(\sum_{i} p_{i,t} = 1\Big)
\end{aligned}$$

Proof of Theorem 3 (bandit)

We highlight the differences from the full-information case.

Step 1: introduce a measure of progress (log-sum-exp)

$$\Phi_t = \frac{1}{\eta} \log \left(\sum_{i=1}^{n} \exp(\eta \widetilde{S}_{i,t}) \right)$$

Basic facts:

- $\Phi_0 = \frac{1}{\eta} \log n$.
- $\Phi_T = \frac{1}{\eta} \log \left(\sum_{i=1}^{n} \exp(\eta \widetilde{S}_{i,T}) \right) \geq \widetilde{S}_{i,T}, \qquad i \in [n]$
- $\mathbb{E}[\Phi_T] \geq \mathbb{E}[\widetilde{S}_{i,T}] = S_{i,T}$.

Proof of Theorem 3 (bandit)

Step 2: understand the temporal difference of Φ_t.

$$\begin{aligned}
\Phi_t - \Phi_{t-1} &= \frac{1}{\eta} \log \left(\sum_{i=1}^{n} e^{\eta \widetilde{S}_{i,t}} \right) - \frac{1}{\eta} \log \left(\sum_{i=1}^{n} e^{\eta \widetilde{S}_{i,t-1}} \right) \\
&= \frac{1}{\eta} \log \frac{\sum_{i=1}^{n} e^{\eta \widetilde{S}_{i,t-1}} e^{\eta \widetilde{r}_{i,t}}}{\sum_{i=1}^{n} e^{\eta \widetilde{S}_{i,t-1}}} \\
&= \frac{1}{\eta} \log \sum_{i=1}^{n} \frac{e^{\eta \widetilde{S}_{i,t-1}}}{\sum_{i=1}^{n} e^{\eta \widetilde{S}_{i,t-1}}} e^{\eta \widetilde{r}_{i,t}} \\
&= \frac{1}{\eta} \log \sum_{i=1}^{n} p_{i,t} e^{\eta \widetilde{r}_{i,t}}
\end{aligned}$$

By same arguments, we obtain

$$\Phi_t - \Phi_{t-1} \leq \sum_{i=1}^{n} p_{i,t} \widetilde{r}_{i,t} + \eta \sum_{i=1}^{n} p_{i,t} \widetilde{r}_{i,t}^{\,2}.$$

Proof of Theorem 3 (bandit)

Step 3: telescoping

$$\Phi_T - \Phi_0 \leq \sum_{t=1}^{T} \sum_{i=1}^{n} p_{i,t} \widetilde{r}_{i,t} + \eta \sum_{t=1}^{T} \sum_{i=1}^{n} p_{i,t} {\widetilde{r}_{i,t}}^2,$$

which leads to

$$\mathbb{E}[\Phi_T] - \Phi_0 \leq \sum_{t=1}^{T} \sum_{i=1}^{n} p_{i,t} r_{i,t} + \eta \sum_{t=1}^{T} \sum_{i=1}^{n} r_{i,t}^2$$

after taking expectations using

$$\mathbb{E}[\widetilde{r}_{i,t}] = r_{i,t} \qquad \mathbb{E}[{\widetilde{r}_{i,t}}^2] = \frac{r_{i,t}^2}{p_{i,t}}.$$

Proof of Theorem 3 (bandit)

Step 4: finishing up.

$$
\begin{aligned}
R_T(r) &= \max_{1 \le i \le n} S_{i,T} - \sum_{t=1}^{T} \sum_{i=1}^{n} p_{i,t} r_{i,t} \\
&\le \mathbb{E}[\Phi_T] - \sum_{t=1}^{T} \sum_{i=1}^{n} p_{i,t} r_{i,t} \\
&\le \Phi_0 + \eta \sum_{t=1}^{T} \sum_{i=1}^{n} r_{i,t}^2 \\
&\le \frac{\log n}{\eta} + \eta n T.
\end{aligned}
$$

Motivation

So far, we have seen that for both stochastic bandits and adversarial bandits, the worst-case regret bound R_T scales as (ignoring logarithmic factors)

$$\widetilde{O}(\sqrt{T}).$$

Question

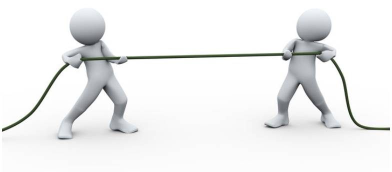
Can we improve the worst-case regret, say to $\widetilde{O}(T^{1/4})$ or $\widetilde{O}(T^{1/3})$?

Two paths:

- Try hard to come up with a better algorithm.
- Develop negative results that show this is impossible. Our plan!

Why studying lower bounds?

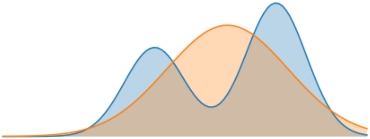
- Lower bounds tell us the minimal price we need to pay.
- Benchmark performance: given an upper bound for certain algorithm, how much room can we improve?
- Matching upper and lower bounds tells us the **fundamental limits**.



Warm-up: basic tools

Which distributions do the data come from?

- Consider hypothesis testing.
- Under different hypotheses, we collect data with different distributions
- The (in)ability to distinguish these distributions becomes the key



Question: how do we measure the distance between distributions?

Distance of distributions

Definition 7 (TV distance)

For two probabilities p, q over Ω, the total variation distance is given by

$$d_{\mathsf{TV}}(p,q) = \sup_{A \subseteq \Omega} p(A) - q(A) \in [0,1].$$

Definition 8 (KL divergence)

For two probabilities p, q over Ω, the Kullback-Leibler (KL) divergence is given by

$$\mathsf{KL}(p\|q) = \sum_{x \in \Omega} p(x) \log \frac{p(x)}{q(x)}.$$

Examples of KL divergence

- **Bernoulli distributions:**

$$\mathsf{KL}\big(\mathsf{Bern}(\tfrac{1+\epsilon}{2})\|\mathsf{Bern}(\tfrac{1}{2})\big) = \frac{1+\epsilon}{2}\log\big(1+\epsilon\big) + \frac{1-\epsilon}{2}\log\big(1-\epsilon\big)$$
$$\le 2\epsilon^2,$$

where the inequality follows from $\log(1+x) \le x$. Note that

$$\mathsf{KL}\big(\mathsf{Bern}(\tfrac{1}{2})\|\mathsf{Bern}(\tfrac{1+\epsilon}{2})\big) \neq \mathsf{KL}\big(\mathsf{Bern}(\tfrac{1+\epsilon}{2})\|\mathsf{Bern}(\tfrac{1}{2})\big)!$$

- **Gaussian distributions:**

$$\mathsf{KL}\big(\mathcal{N}(\mu_1,\sigma^2)\|\mathcal{N}(\mu_2,\sigma^2)\big) = \frac{(\mu_1-\mu_2)^2}{2\sigma^2}$$

Pinsker's inequality

Pinsker's inequality

For two probability distributions p, q over Ω, it holds that

$$2d_{\mathsf{TV}}(p,q)^2 \leq \mathsf{KL}(p\|q).$$

- By definition, for any event $A \subseteq \Omega$,

$$p(A) - q(A) \leq \sqrt{\frac{1}{2}\mathsf{KL}(p\|q)}.$$

- Due to asymmetry of KL divergence, we have:

$$2d_{\mathsf{TV}}(p,q)^2 \leq \min\left\{\mathsf{KL}(p\|q), \mathsf{KL}(p\|q)\right\}.$$

- A very useful tool!

Toy example: binary hypothesis testing

Suppose we observe a sequence of coin flips

$$A_t \overset{\text{i.i.d.}}{\sim} \text{Ber}(\mu), \quad t = 1, \ldots, T.$$

Consider two hypotheses for μ:

$$\mathcal{H}_0 : \quad \mu = \frac{1}{2}, \qquad \mathcal{H}_1 : \quad \mu = \frac{1+\epsilon}{2}.$$

We want to determine which hypothesis is true. Is the coin fair?

Question

How many samples do we need to collect in order to do so reliably?

Step 1: KL divergence of the data

Denote the data distribution under two hypotheses respectively as

$$\mathbb{P}_0 := \mathbb{P}\left(A_1, A_2 \ldots, A_T | \mathcal{H}_0\right)$$
$$\mathbb{P}_1 := \mathbb{P}\left(A_1, A_2 \ldots, A_T | \mathcal{H}_1\right).$$

Then, it is easy to see

$$\begin{aligned} \mathsf{KL}(\mathbb{P}_1 \| \mathbb{P}_0) &= \sum_{i=1}^{T} \mathsf{KL}\left(\mathbb{P}(A_i|\mathcal{H}_1) \| \mathbb{P}(A_i|\mathcal{H}_0)\right) \\ &= T \cdot \mathsf{KL}\big(\mathsf{Bern}(\tfrac{1+\epsilon}{2}) \| \mathsf{Bern}(\tfrac{1}{2})\big) \\ &\leq 2T\epsilon^2. \end{aligned}$$

The KL divergence scales linear in T and quadratically in ϵ.

Step 2: determine the goal

Question: what do we mean by solving the problem "reliably"?

Answer: Maybe getting a correct answer with non-trivial probability, e.g. for some small probability of error δ,

$$\mathbb{P}\left(\text{learner outputs fair}|\mathcal{H}_0\right) \geq 1 - \delta/2.$$
$$\mathbb{P}\left(\text{learner outputs unfair}|\mathcal{H}_1\right) \geq 1 - \delta/2.$$

Let us call the event $A = \{\text{learner outputs fair}\}$, then the above leads to

$$\mathbb{P}_0(A) \geq 1 - \delta/2, \qquad \mathbb{P}_1(A) \leq \delta/2$$

$$\Longrightarrow \qquad \mathbb{P}_0(A) - \mathbb{P}_1(A) \geq 1 - \delta.$$

Step 3: applying Pinsker

By Pinsker's inequality, we know

$$2(\mathbb{P}_0(A) - \mathbb{P}_1(A))^2 \leq \mathsf{KL}(\mathbb{P}_1 \| \mathbb{P}_2) \leq 2T\epsilon^2.$$

Hence,

$$\begin{aligned} T &\geq \frac{(\mathbb{P}_0(A) - \mathbb{P}_1(A))^2}{\epsilon^2} \\ &\geq \frac{(1-\delta)^2}{\epsilon^2}. \end{aligned}$$

The sample size T needs to be at least

$$T \gtrsim \frac{1}{\epsilon^2}!$$

Examine the upper bound

The scaling $T \gtrsim \frac{1}{\epsilon^2}$ turns out to be sufficient too!

By Hoeffding's inequality, we know

$$\left| \frac{1}{n} \sum_{t=1}^{T} A_t - \mu \right| \le \sqrt{\frac{\log(2/\delta)}{2T}} \qquad \text{with prob. } 1 - \delta.$$

By setting $\sqrt{\frac{\log(2/\delta)}{2T}} \le \frac{\epsilon}{4}$, or equivalently, $T \ge \frac{8\log(2/\delta)}{\epsilon^2}$, we guarantee

$$\left| \frac{1}{n} \sum_{t=1}^{T} A_t - \mu \right| \le \frac{\epsilon}{4} \qquad \text{with prob. } 1 - \delta.$$

The learner compares the sample mean $\frac{1}{n}\sum_{t=1}^{T} A_t$ with $\frac{1}{2} + \frac{\epsilon}{4}$.

Lower bounds for multi-arm bandits

Worst-case lower bound

For simplicity, we will assume all arms have a Gaussian reward distribution $\mathcal{N}(\mu_i, 1)$ for $i \in [n]$.

Theorem 9 (minimax lower bound)

Let $n > 1$ and $T \geq n-1$. Then, for any algorithm π, there exists a mean vector $\mu = [\mu_i]_{1 \leq i \leq n} \in [0,1]^n$ such that

$$R_T \geq \frac{1}{27}\sqrt{(n-1)T}.$$

- No algorithm can obtain a regret bound better than $\Omega(\sqrt{T})$.
- Stochastic bandits are easier than adversarial bandits. Lower bounds for stochastic bandits are also applicable for adversarial bandits.
- Certifies the near-optimality of $\sqrt{T}$ regret for UCB [Auer et al., 2002a] and EXP3 [Auer et al., 2002b].

Instance-dependent lower bound

[Lai and Robbins, 1985]: we might be able to say something less pessimistic in an instance-dependent manner.

Theorem 10 (Instance-dependent lower bound)

Consider a strategy that satisfies $\mathbb{E}[R_T] = o(T^\alpha)$ for any set of reward distributions $\{\mathbb{P}_i\}_{1 \le i \le n}$ indexed by a single real parameter, any arm i with sub-optimality gap $\Delta_i > 0$, and any $\alpha > 0$. Then, the following holds

$$\liminf_{T \to \infty} \frac{R_T}{\log T} \ge \sum_{i:\ \Delta_i > 0} \frac{\Delta_i}{\mathsf{KL}(\mathbb{P}_i \| \mathbb{P}^\star)},$$

where $\mathbb{P}^\star$ is the distribution of the optimal arm.

- The instance-dependent lower bound is $\Omega(\log T)$.

Near instance-optimality of UCB

For Gaussian bandits,

$$\mathsf{KL}(\mu_i \| \mu^\star) = \frac{\Delta_i^2}{2},$$

then it follows that

$$\liminf_{T\to\infty} \frac{R_T}{\log T} \geq \sum_{i:\ \Delta_i > 0} \frac{2}{\Delta_i},$$

and

$$R_T \gtrsim \sum_{i:\ \Delta_i > 0} \frac{\log T}{\Delta_i}.$$

- This almost matches with the instance-dependent upper bound of UCB, which says (ignoring n)

$$R_T \lesssim \sum_{i:\Delta_i > 0} \frac{\log T}{\Delta_i}.$$

Analysis

KL Divergence between two bandits

Lemma 11 (Divergence decomposition)

- *Let $\nu = (\mathbb{P}_1, \ldots, \mathbb{P}_n)$ be the reward distributions associated with one n-armed bandit, and let $\nu' = (\mathbb{P}'_1, \ldots, \mathbb{P}'_n)$ be the reward distributions associated with another n-armed bandit.*
- *Fix some algorithm π and let $\mathbb{P}_\nu = \mathbb{P}_{\nu\pi}$ and $\mathbb{P}_{\nu'} = \mathbb{P}_{\nu'\pi}$ be the probability measures on the bandit model $\{i_t, r_t\}_{t=1}^T$ induced by the T-round interconnection of π and ν (respectively, π and ν').*

Then,

$$\mathsf{KL}(\mathbb{P}_\nu \| \mathbb{P}_{\nu'}) = \sum_{i=1}^n \mathbb{E}_\nu[T_{i,T}] \mathsf{KL}(\mathbb{P}_i \| \mathbb{P}'_i),$$

where $T_{i,T} = \sum_{t=1}^T \mathbb{I}(i_t = i)$.

Bretagnolle-Huber inequality

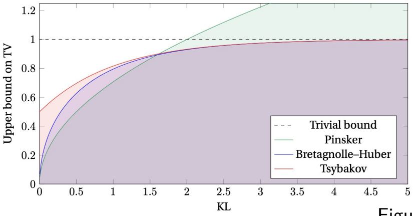


Figure credit: [Canonne, 2022].

Theorem 12 (Bretagnolle-Huber inequality)

For two probability distributions p, q over Ω, it follows that

$$d_{\mathsf{TV}}(p,q) \le \sqrt{1 - e^{-\mathsf{KL}(p\|q)}} \le 1 - \frac{1}{2}e^{-\mathsf{KL}(p\|q)}$$

- The second bound is due to [Tsybakov, 2008].
- As a consequence, for any event $A \subseteq \Omega$,

$$p(A^c) + q(A) \ge \frac{1}{2}e^{-\mathsf{KL}(p\|q)}.$$

Proof of minimax lower bound

Step 1: identifying a pair of bandits. Fix an algorithm π.

Suppose we begin with a Gaussian bandit ν with unit variance $\mathbb{P}_i = \mathcal{N}(\mu_i, 1)$, where $\mu^\star = \mu_1 > \mu_2 \geq \ldots \geq \mu_n$ w.l.o.g.. Let $\mathbb{P}_\nu$ be the resulting probability measure over T-round interconnection of π and ν.

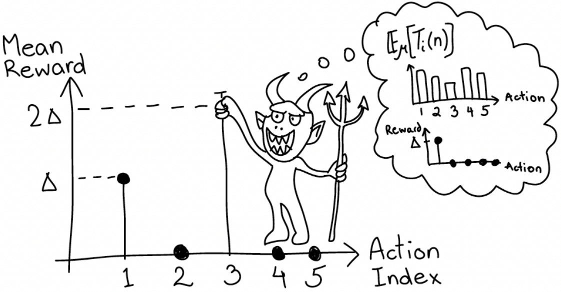


Figure 15.1 The idea of the minimax lower bound. Given a policy and one environment, the evil antagonist picks another environment so that the policy will suffer a large regret in at least one environment.

Figure credit: [Lattimore and Szepesvári, 2020]

Proof of minimax lower bound

Identifying the competing bandit: in view of the divergence decomposition lemma, let

$$j = \arg\min_{i>1} \mathbb{E}_\nu[T_{i,T}],$$

the arm that has been least pulled. The second bandit ν' is then selected as

$$\mathbb{P}'_i = \begin{cases} \mathbb{P}_i & i \neq j \\ \mathcal{N}(\mu_j + \lambda, 1) & i = j \end{cases},$$

where $\lambda > \Delta_j$ is to be selected. Arm j is optimal in the second bandit. Call the resulting probability measure $\mathbb{P}_{\nu'}$.

Proof of minimax lower bound

Step 2: computing the KL divergence. It is easy to observe that

$$\mathsf{KL}(\mathbb{P}_\nu \| \mathbb{P}_{\nu'}) = \mathbb{E}_\nu[T_{j,T}]\mathsf{KL}(\mathbb{P}_j \| \mathbb{P}'_j) \le \frac{T\lambda^2}{2(n-1)},$$

where we used

$$\sum_{i>1} \mathbb{E}_\nu[T_{i,T}] \le \sum_{i=1}^{n} \mathbb{E}_\nu[T_{i,T}] = T \qquad \Longrightarrow \qquad \mathbb{E}_\nu[T_{j,T}] \le \frac{T}{n-1}.$$

and

$$\mathsf{KL}\big(\mathcal{N}(\mu_j, 1) \| \mathcal{N}(\mu_j + \lambda, 1)\big) = \frac{\lambda^2}{2}.$$

Proof of minimax lower bound

Step 3: summing up the regrets of two bandits. By the regret decomposition lemma,

- For the first bandit ν, since j is sub-optimal,

$$R_T = \sum_{i \neq 1} \Delta_i \mathbb{E}_\nu[T_{i,T}] \geq \Delta_j \mathbb{E}_\nu[T_{j,T}] \geq \frac{T\Delta_j}{2} \mathbb{P}_\nu(T_{j,T} \geq T/2).$$

- For the second bandit, since j is optimal, for any $i \neq j$, it follows $\Delta'_i = \mu_j + \lambda - \mu_i = \lambda - (\mu_i - \mu_j) \geq \lambda - \Delta_j$, it follows

$$R'_T = \sum_{i \neq j} \Delta'_i \mathbb{E}_{\nu'}[T_{i,T}] \geq \frac{T(\lambda - \Delta_j)}{2} \mathbb{P}_{\nu'}(T_{j,T} < T/2).$$

Letting $A = \{T_{j,T} < T/2\}$, and summing these up, we have

$$R_T + R'_T \geq \frac{T}{2} \min\{\Delta_j, \lambda - \Delta_j\} \left[\mathbb{P}_\nu(A) + \mathbb{P}_{\nu'}(A^c)\right].$$

Proof of minimax lower bound

Step 4: finishing up by Bretagnolle-Huber. By Bretagnolle-Huber,

$$\mathbb{P}_\nu(A) + \mathbb{P}_{\nu'}(A^c) \geq \frac{1}{2} e^{-\mathsf{KL}(\mathbb{P}_\nu \| \mathbb{P}_{\nu'})} \geq \frac{1}{2} \exp\left(-\frac{2T\lambda^2}{(n-1)}\right).$$

$$\Longrightarrow \qquad R_T + R'_T \geq \frac{T}{4} \min\left\{\Delta_j, \lambda - \Delta_j\right\} \exp\left(-\frac{T\lambda^2}{2(n-1)}\right).$$

Setting $\lambda = 2\Delta_j$ leads to

$$R_T + R'_T \geq \frac{T}{4} \Delta_j \exp\left(-\frac{2T\Delta_j^2}{(n-1)}\right).$$

Let $\mu^\star = \mu_1 = \Delta$ and $\mu_2, \ldots, \mu_n = 0$. Set $\Delta_j = \Delta = \sqrt{(n-1)/4T} \leq 1/2$, we have

$$R_T + R'_T \geq \frac{e^{-1/2}}{8} \sqrt{(n-1)T} \implies \max\{R_T,\, R'_T\} \geq \frac{e^{-1/2}}{16} \sqrt{(n-1)T}.$$

Proof of instance-dependent lower bound

We only consider **Gaussian bandits**.

In view of the regret decomposition lemma, it is sufficient to show for any sub-optimal arm i,

$$\operatorname{liminf}_{T\to\infty} \frac{\mathbb{E}_\nu[T_{i,T}]}{\log T} \geq \frac{2}{\Delta_i^2}.$$

Let us fix a sub-optimal arm $j \neq i^\star$.

Step 1: identify the competing bandit. Motivated to the earlier proof, we construct second bandit ν' is then selected as

$$\mathbb{P}'_i = \begin{cases} \mathbb{P}_i & i \neq j \\ \mathcal{N}(\mu_j + \lambda, 1) & i = j \end{cases},$$

where we set $\lambda > \Delta_j$, and arm j is optimal in the second bandit.

Proof of instance-dependent lower bound

Step 2: lower bound the regret via Bretagnolle-Huber. Similar to the earlier proof, we obtain

$$
\begin{aligned}
R_T + R'_T &\geq \frac{T \min\{\Delta_j, \lambda - \Delta_j\}}{4} e^{-\mathsf{KL}(\mathbb{P}_\nu \| \mathbb{P}_{\nu'})} \\
&= \frac{T \min\{\Delta_j, \lambda - \Delta_j\}}{4} e^{-\lambda^2 \mathbb{E}_\nu[T_{j,T}]/2},
\end{aligned}
$$

which gives

$$
\begin{aligned}
& \mathbb{E}_\nu[T_{j,T}] \geq \frac{2}{\lambda^2} \log\left(\frac{T \min\{\Delta_j, \lambda - \Delta_j\}}{4(R_T + R'_T)}\right) \\
\Longrightarrow \quad & \frac{\lambda^2}{2} \frac{\mathbb{E}_\nu[T_{j,T}]}{\log T} \geq \left[1 + \frac{\log \min\{\Delta_j, \lambda - \Delta_j\}}{4 \log T} - \frac{\log(R_T + R'_T)}{\log T}\right].
\end{aligned}
$$

The next step is to examine the liminf of the right-hand-side.

Proof of instance-dependent lower bound

Step 3: taking limits to finish up. Since for any $\alpha > 0$, there exist some constant $C_\alpha > 0$ such that

$$R_T + R'_T \leq C_\alpha T^\alpha$$

for all T, we have

$$\mathsf{limsup}_{T\to\infty} \frac{\log(R_T + R'_T)}{\log T} \leq \mathsf{limsup}_{T\to\infty} \frac{\alpha \log T + \log C_\alpha}{\log T} = \alpha.$$

Since this holds for any arbitrary $\alpha > 0$, it follows that

$$\mathsf{limsup}_{T\to\infty} \frac{\log(R_T + R'_T)}{\log T} = 0.$$

Consequently,

$$\mathsf{liminf}_{T\to\infty} \frac{\lambda^2}{2} \frac{\mathbb{E}_\nu[T_{j,T}]}{\log T} \geq 1.$$

Taking the infimum of both sides over $\lambda > \Delta_j$ thus finishes the proof.

Further pointers

The literature on bandits is vast, and we have only scratched the surface.

Further pointers to worthy topics:

- Thompson sampling: a Bayesian approach
- Beyond EXP3: dealing with variance

Excellent reference: Bandit algorithms [Lattimore and Szepesvári, 2020].

References I

Arora, S., Hazan, E., and Kale, S. (2012).
The multiplicative weights update method: a meta-algorithm and applications.
Theory of Computing, 8(1):121–164.

Auer, P., Cesa-Bianchi, N., and Fischer, P. (2002a).
Finite-time analysis of the multiarmed bandit problem.
Machine learning, 47(2):235–256.

Auer, P., Cesa-Bianchi, N., Freund, Y., and Schapire, R. E. (2002b).
The nonstochastic multiarmed bandit problem.
SIAM Journal on Computing, 32(1):48–77.

Canonne, C. L. (2022).
A short note on an inequality between KL and TV.
arXiv preprint arXiv:2202.07198.

Lai, T. L. and Robbins, H. (1985).
Asymptotically efficient adaptive allocation rules.
Advances in Applied Mathematics, 6(1):4–22.

Lattimore, T. and Szepesvári, C. (2020).
Bandit algorithms.
Cambridge University Press.

References II

Littlestone, N. and Warmuth, M. K. (1994).
The weighted majority algorithm.
Information and Computation, 108(2):212–261.

Tsybakov, A. B. (2008).
Introduction to Nonparametric Estimation.
Springer Publishing Company, Incorporated, 1st edition.

Vovk, V. G. (1990).
Aggregating strategies.
In *Proceedings of the third annual workshop on Computational learning theory*, pages 371–386.